

The Pipes and the Rents

Value capture, structural moats, and margin asymmetry between platform aggregators and telecom infrastructure in India's digital economy

A strategy case study · Ayaan Nawaz

Executive summary

The claim that “platforms win” is true but loosely stated, and the looseness is where it gets attacked. India's telecom industry earns far *more* revenue than its digital-advertising market: combined wireless revenue runs to roughly ₹2.7 lakh crore a year³, while digital advertising — the platforms' core monetisation engine — is only now approaching ₹1 lakh crore (about 60–64% of all ad spend).^{1,2} So “winning” cannot mean size. The defensible claim is narrower and sharper: **per rupee of capital deployed, the aggregation layer of the digital stack captures disproportionately more economic profit than the infrastructure layer beneath it or the creator layer above it.** Alphabet earns a return on invested capital near 24.6%;⁷ Bharti Airtel, after a decade-long capital-expenditure super-cycle, earns about 14% — far better than its 6.4% trough in 2022,⁶ but only modestly above its cost of capital. That gap, not the revenue gap, is the story this study defends, tests against the obvious counterexample, and then tries to break.

1. The question hidden inside “platforms win”

The intuitive comparison is between companies: Google versus Airtel, the asset-light monetiser versus the asset-heavy builder. It is a tidy contrast and it is the wrong unit of analysis, because the single largest Indian telecom refutes it in one move. Reliance Jio built the most expensive network in Indian history *and* the most-used consumer app stack on top of it. If the frame is “telecoms are pipes and platforms are aggregators,” Jio is neither and both.

The fix is to stop comparing firms and start comparing **layers of the stack**. There are three: *infrastructure* (spectrum, towers, fibre — the physical capacity to move bits); *aggregation* (search, social feeds, marketplaces — the software that sits between supply and demand and owns the user relationship); and *creation* (the content and goods that flow through). The thesis of this study is about where in that stack the rents collect, irrespective of which corporate entity happens to own each layer. Framed this way, Jio stops being a contradiction and becomes the most instructive case in the entire analysis — which is exactly why it gets its own section.

2. The lens: aggregation theory

The organising framework is aggregation theory. Its core observation is that when supply becomes abundant and distribution becomes nearly free, power migrates from whoever controls supply to whoever controls demand. The firm that owns the user relationship can commoditise the suppliers behind it, because it — not they — decides what the user sees. A telecom can lay the most fibre in the country and still find that the economic surplus flows past it to the handful of destinations its customers actually open. The pipe is necessary; necessity is not the same as pricing power.

This lens makes three testable predictions, which form the three tests below: that the aggregator's pricing power is structurally different in kind from the telecom's; that the margin asymmetry is real but shows up in return on capital rather than in headline operating margin; and that the durable moat is built on demand and data, not on capital deployed.

3. Test one — pricing power, and a correction

The popular version of this argument holds that telecoms have no pricing power because data is a commodity, so a price rise by one operator simply pushes customers to the next. That was a fair description of India in 2018. It is no longer true, and a case study that repeats it forfeits its credibility on the first cross-examination.

After a brutal consolidation, the market is now a three-player oligopoly — Jio at roughly 46.5% subscriber share, Airtel near 36.6%, Vodafone Idea around 17%.³ The July 2024 tariff hike stuck across all three rather than triggering an exodus, and analysts expect a further 16–20% round in 2026 that would carry Airtel's ARPU from about ₹256 today toward ₹300 by FY27.⁴ That is oligopoly pricing power, not its absence.

So the honest contrast is not “platforms have pricing power and telecoms don't.” It is the **kind** of pricing power. The aggregator's is auction-cleared, granular, and rises automatically with demand: every additional advertiser bidding

on the same query lifts the clearing price without the platform lifting a finger. The telecom's is coarse, infrequent, politically scrutinised, and ultimately capped by household affordability. Both can raise prices; only one of them does so continuously and without friction. Stating it this way *strengthens* the thesis rather than overstating it.

4. Test two — margins and the return-on-capital asymmetry

Here too the naive version misleads. It is not true that telecom operating margins are thin: Airtel's EBITDA margin sits around 50–57%,⁵ comparable to or above many platforms' *operating* margins. If the comparison stopped at EBITDA, the asset-light story would look wrong on its face.

The asymmetry only appears *below* the EBITDA line. Depreciation on the network and the relentless reinvestment in spectrum and fibre consume the cash that EBITDA appears to generate, which is why the honest metric is return on invested capital — profit measured against every rupee the business has had to sink in to earn it. On that measure the gap is wide and persistent.

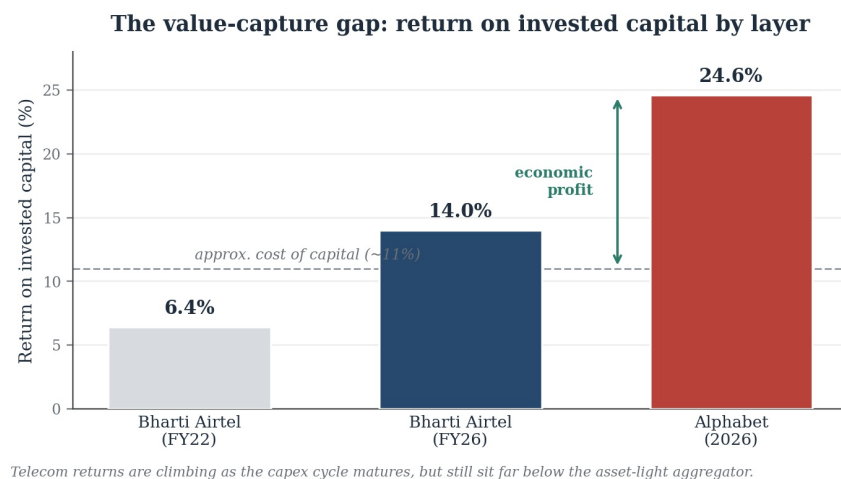


Figure 1. Return on invested capital. The telecom's returns have roughly doubled off their 2022 trough as the capex cycle matures and tariffs rise — an important nuance — yet still sit only modestly above the cost of capital, while the asset-light aggregator earns roughly twice as much per rupee invested. The green span is economic profit: return earned in excess of the cost of capital.

Two caveats keep this honest. First, telecom returns are *improving*, not collapsing; the FY22-to-FY26 climb is real and the trend matters. Second, the headline gap is somewhat overstated by accounting: platforms expense most of their R&D and customer-acquisition spend rather than capitalising it, which understates their invested capital and flatters their ROIC. Capitalise those intangibles and the true gap narrows — but it does not close.

5. Test three — distribution and the shape of the moat

“Proprietary data” is the phrase every deck reaches for and few decompose. The moat is really three mechanisms working together. **Demand aggregation:** users come to the platform first, so suppliers must follow it on its terms. **Data network effects:** more usage yields better targeting, which lifts ad yield, which funds more product, which attracts more usage — a loop the telecom cannot enter because it sees traffic, not intent. **Switching and multi-homing costs:** the identity, history and habit live with the platform, not the pipe.

The structural point is that a user reaches Google *through* Airtel, but everything monetisable about that user — what they searched, what they want, what they will click — accrues to Google. The telecom owns a toll road; the platform owns the destination. Toll roads earn a regulated, capital-intensive return. Destinations earn rents.

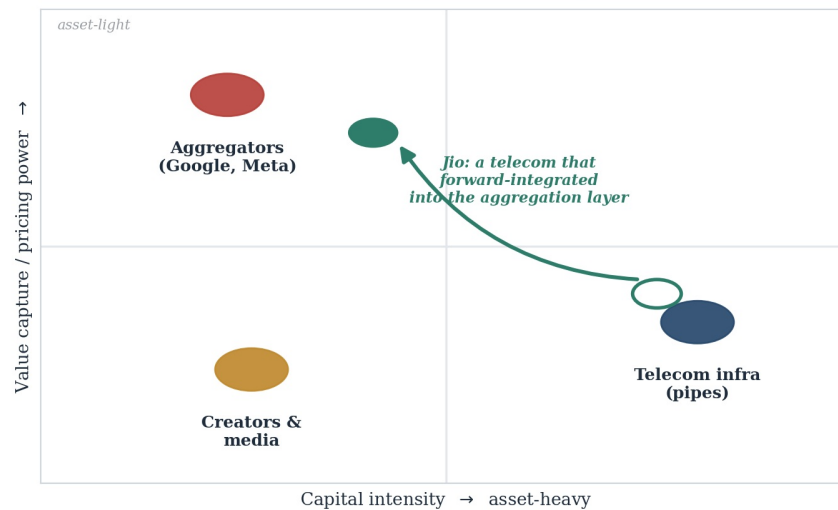


Figure 2. The digital stack mapped on capital intensity against value capture. Aggregators sit asset-light and rent-rich; pure infrastructure sits capital-heavy and return-constrained; creators sit low and volatile. The green arrow is the move that defines the next decade — a telecom climbing out of the infrastructure quadrant into the aggregation layer.

6. The objection that should break the thesis — Jio

If the thesis is “telecoms carry the capex and platforms capture the value,” then the largest Indian telecom is its cleanest refutation. Reliance did not stay a pipe. On top of the network it built MyJio, the direct-to-consumer hub with more than 300 million monthly active users;⁸ JioMart in commerce; JioCinema and JioSaavn in content; JioFinance in payments; an advertising stack; and, through the parent, ownership of a major media house. Jio's digital-services revenue reached about ₹16,500 crore in 2024, up 15% year on year.⁸ This is not a pipe that happens to run some apps; it is a deliberate, conglomerate-scale attempt to occupy the aggregation layer.

WHY THIS DOES NOT SINK THE THESIS — IT CONFIRMS THE REFRAMING

Value still accrues to the aggregation layer. Jio's strategic insight was precisely that remaining infrastructure was a losing position, so it used the pipe to bootstrap the aggregator and capture the layer above. The lesson is not “telecoms win after all.” It is that the boundary between infrastructure and aggregation is not a law of nature — it is a position firms can attack, at enormous cost. Airtel is now attempting a lighter version of the same climb with Airtel Ads and its content bundles. The open question for the next decade is whether infrastructure players can ascend the stack faster than aggregators can commoditise the pipe beneath them.

7. What could still be wrong

Following the discipline of any honest causal claim, each threat below is stated *with the direction it pushes the conclusion*, not just named.

Forward integration

Direction: erodes the thesis. If telecoms routinely climb into aggregation, the layer's rents leak back toward infrastructure owners. So far only Reliance has managed it, and it required an entire conglomerate and a once-in-a-generation balance sheet to do so — which is itself evidence of how defended the aggregation layer is.

ONDC — state-sponsored de-aggregation

Direction: dissolves the moat at its root. The government is explicitly trying to do for commerce what UPI did for payments — replace the proprietary aggregator with an open protocol so that the buyer app no longer owns the user. By early 2025 the network counted roughly 3.5 lakh sellers and on the order of 1.2 crore monthly transactions;⁹ real, but still a sliver of platform volumes, and still wrestling with trust, fulfilment and user-experience fragmentation. If it works it is the single largest threat to the thesis; whether it works is genuinely unproven.

Regulation — CCI and the DPDP Act

Direction: narrows the moat at the edges. The Competition Commission's ₹1,337.76 crore Android penalty (upheld on appeal in 2023)¹⁰ and an earlier data-related fine, together with the draft Digital Competition Bill and the Digital Personal Data Protection Act of 2023,¹¹ all take aim at the gatekeeper position and the data advantage directly. Realistically this raises the cost of holding the moat rather than reversing the economics, but it is a structural headwind that did not exist five years ago.

Direction: redistributes the rents *within* the aggregation layer rather than out of it. Amazon, Flipkart and quick-commerce are pulling advertising toward transaction-rich environments, fragmenting the search-and-social duopoly; and generative AI threatens the search box itself, the original aggregation chokepoint. The layer still wins — but the names that own it may change.

8. What this study cannot claim

A return-on-capital comparison between a US-listed platform and an India-listed telecom mixes currencies, tax regimes and accounting conventions; a cleaner version normalises for these and capitalises intangibles, which would shrink the headline gap (though not erase it). The pricing-power and moat arguments rest on public disclosures and secondary research, not on internal unit economics. And aggregation theory is a lens, not a proof: it organises the facts persuasively, but unlike a regression coefficient it is not cleanly falsifiable. The right next step would pair this with a quantified economic-profit decomposition of the Indian digital stack using normalised invested capital — the equivalent of moving from a 2×2 to a full panel.

9. What we actually learned

On substance: in India's digital stack the rents collect in the aggregation layer; the asymmetry is invisible at the revenue line, invisible even at EBITDA, and only resolves into focus at return on invested capital; and the moat that protects it is demand-side and data-side, not capital-side. **On method:** the work was never the assertion that platforms win — that is a headline anyone can write. The work was specifying the claim precisely enough to be wrong (per rupee of capital, not in absolute size), confronting the one case that should have broken it (Jio), and signing the direction of every force that could still overturn it. A red-amber-green matrix hands you a verdict. This hands you a verdict you can defend under questioning — and tells you honestly which parts you have not yet earned.

Summary of claims

Claim	Evidence / estimate	How much to trust it
The aggregation layer captures disproportionate value per unit of capital	ROIC ~24.6% vs ~14%	Strong on direction — gap somewhat overstated by intangible accounting
Telecom pricing power is rising, not absent — the contrast is kind, not presence	ARPU ₹256 → ~₹300 by FY27; industry-wide 2024 hike held	Strong — analyst consensus, corrects a common error
The moat is demand-side (aggregation + data), not capital-side	Qualitative, aggregation-theory framework	Solid as a lens — persuasive but not falsifiable
The thesis survives the Jio counterexample	Reframed from firm to layer	Solid — but the layer boundary is contestable
ONDC / regulation could dissolve the moat	~1.2 cr ONDC txns/month; CCI & DPDP actions	Weakest — live experiment, outcome unproven

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Note. Figures are compiled from public reporting and brokerage research and are approximate — used to establish direction and order of magnitude, not audited precision. Live market metrics (ARPU, ROIC) reflect the most recent values available as of mid-2026 and will drift over time.